

THE THRONES OF IBD

THE THRONE ROOM

ASSASSIN 1
BLOCK · 55%

ASSASSIN 2
BIND · 35%

ASSASSIN 3
JUICE · 80%

ASSASSIN 4
ANTI-PC · 90%

THE PORTRAIT GALLERY

DEMOGRAPHICS WALL

LARGE PORTRAIT
70-80 · 120/100K

SMALLER PORTRAITS

THYROID

VITILIGO

T1DM

PANCREAS

HYPOPARATHYROID

HYPOGAMMA

CLINICAL

GROUP A · F>M 1.6:1

- GRAY HAIR
- BLUE EYES
- HLA-B8
- FAMILY HX↑

AFRICAN
EARLIER ONSET

LATINO
EARLIER

ASIAN
LESS COMMON

PALE

YELLOW

RED GLOSSY TONGUE

easy BRUISES

CRACKED CORNERS

CRREED

ASYMPTOMATIC

CBC
↑MCV

FOLATE: NO NEURO

50% ENDOCRINE

JUVENILE PA

CANDIDIASIS

CONGENITAL IF

- AR · AGE 1-3
- NORMAL MUCOSA
- NO ANTIBODIES

GASTRIC CA
↑MEN · LIFE EXPECTANCY
↓MALES

1 SCOPE

H. PYLORI

CHRONIC GASTRITIS

IDA

PA

CRUMBLING

ASSASSIN 4
H⁺/K⁺ ATPase

PARIENTAL CELL REPLACED BY MUCOUS

ATROPHY

INTESTINAL METAPLASIA

ANTRUM PRESERVED

GRABBING SLAMPLEX

CELL-MEDIATED → IMMUNITY

ILEUM

BETWEEN

POURING ACID

SMASHING

CRUMBLING

DECAYING THRONE

IF

1. Autoimmune atrophic gastritis

WHAT YOU SEE

THRONE ROOM 'DECAYING THRONE', 'HYPOCHLORHYDRIA HYPERGASTRINEMIA'

WHAT IT ENCODES

Pernicious anemia (PA) is autoimmune metaplastic atrophic gastritis of the gastric BODY/FUNDUS: T-cell-mediated destruction of oxyntic (parietal) cells eliminates both intrinsic factor (→ B12 malabsorption) and acid (→ achlorhydria/hypochlorhydria). Loss of acid removes negative feedback on antral G-cells, so gastrin rises markedly (hypergastrinemia is a key serologic clue). Chronic hypergastrinemia drives ECL-cell hyperplasia → type 1 gastric carcinoid risk. Mnemonic: 'no parietal cells → no IF (anemia) + no acid (high gastrin) → carcinoid + cancer risk.'

2. Anti-parietal cell antibody

WHAT YOU SEE

'ASSASSIN' figure 'ANTI-PC 90%', H+/K+ ATPase

WHAT IT ENCODES

Anti-parietal cell antibodies target the H+/K+ ATPase proton pump and are present in ~85-90% of PA — high SENSITIVITY but LOW SPECIFICITY (also seen in atrophic gastritis without PA, Hashimoto thyroiditis, type 1 diabetes, and ~10% of healthy elderly). So a positive anti-PC supports but does not confirm PA. Pair it with the more specific anti-IF antibody. Mnemonic: 'Parietal = Plentiful/sensitive; Intrinsic factor = specifc.'

3. Anti-IF antibody (specific)

WHAT YOU SEE

'ANTI-IF' assassin, IF crown

WHAT IT ENCODES

Anti-intrinsic-factor antibodies are present in only ~50% (less SENSITIVE) but are highly SPECIFIC (>95%) for PA — a positive anti-IF essentially clinches the diagnosis in a B12-deficient patient. Two types: blocking antibodies (prevent B12 binding to IF, ~most common) and binding antibodies (prevent the IF–B12 complex from attaching to ileal cubam receptors). Because of its specificity, anti-IF is the confirmatory test; Schilling test is obsolete. Mnemonic: 'IF antibody = If positive, It's PA.'

BOARD TRAP

Calling anti-IF the 'sensitive' test. It is the SPECIFIC one (high specificity, only ~50% sensitive). Anti-parietal cell is the sensitive (~90%) but nonspecific test. A negative anti-IF does NOT exclude PA — confirm with anti-PC, low B12/high MMA, and endoscopy/gastrin if needed.

4. Cell-mediated immunity / ileum

WHAT YOU SEE

'BIND 35%', grabbing IF, ILEUM

WHAT IT ENCODES

B12 absorption requires IF binding in the duodenum and uptake of the IF–B12 complex by cubam (cubilin–amnionless) receptors in the TERMINAL ILEUM. PA blocks this in two ways: loss of IF secretion (atrophy) plus blocking/binding anti-IF antibodies. This is why PA and ileal disease (Crohn, ileal resection, fish tapeworm *Diphyllobothrium*, blind-loop bacterial overgrowth) all cause B12 deficiency. Practical point: high-dose ORAL B12 still works in PA because ~1% is absorbed passively, bypassing IF.

5. Intestinal metaplasia

WHAT YOU SEE

'INTESTINAL METAPLASIA', atrophy

WHAT IT ENCODES

Long-standing autoimmune atrophy progresses through the Correa cascade: chronic gastritis → atrophy → intestinal metaplasia → dysplasia → adenocarcinoma. Intestinal metaplasia of the gastric corpus mucosa is the histologic marker of this premalignant trajectory and underlies the gastric cancer risk in PA. This is the rationale some give for one-time screening endoscopy at PA diagnosis. Mnemonic: 'atrophy → metaplasia → dysplasia → cancer (the gastric stepwise road).'

6. Antrum spared

WHAT YOU SEE

'ANTRUM PRESERVED'

WHAT IT ENCODES

Autoimmune (type A) gastritis is BODY/FUNDUS-predominant and characteristically SPARES THE ANTRUM, because that is where the parietal/oxyntic cells live. This contrasts with *H. pylori* (type B) gastritis, which is antral-predominant. The spared, intact antral G-cells are exactly why gastrin shoots up in PA (no acid feedback). Mnemonic: 'Autoimmune = Antrum-spared, Acid-low, gAstrin-high; *H. pylori* = antrum-hit.'

7. Gastric cancer risk

WHAT YOU SEE

'GASTRIC CA' crown, '3x life expectancy', males

WHAT IT ENCODES

PA carries roughly a 2-3x increased risk of gastric ADENOCARCINOMA and a markedly increased risk of type 1 gastric CARCINOID (from gastrin-driven ECL hyperplasia). New dyspepsia, iron deficiency, weight loss or bleeding in a PA patient warrants endoscopy. Type 1 gastric carcinoids are usually small, multiple, indolent and managed endoscopically. Mnemonic: 'PA = Premalignant Antrum-sparing gastritis → Adenocarcinoma + cARcinoid.'

8. *H. pylori* chronic gastritis

WHAT YOU SEE

'H. PYLORI CHRONIC GASTRITIS' → IDA → PA

WHAT IT ENCODES

H. pylori chronic gastritis can precede and even trigger autoimmune atrophic gastritis (molecular mimicry against H+/K+ ATPase) and independently causes iron-deficiency anemia and B12 malabsorption via hypochlorhydria. Boards link: an older patient with IDA refractory to oral iron may have atrophic gastritis/PA, and treating *H. pylori* can improve absorption. Mnemonic: '*H. pylori* first damages the stomach; autoimmunity finishes the job.'

9. Demographics 70-80 / 120k

WHAT YOU SEE

PORTRAIT GALLERY large portrait '70-80', '120/100k'

WHAT IT ENCODES

Classic PA patient: older adult (peak ~60-80 yrs), of Northern European or African descent, female slightly more than male; prevalence rises with age. It is the most common cause of B12 deficiency due to malabsorption in adults. A board stem typically pairs an elderly patient with macrocytic anemia, glossitis, and neuro symptoms. Mnemonic: 'PA = Pale, Aged northern-European with a sore tongue and tingling feet.'

10. Associated autoimmune disease

WHAT YOU SEE

Smaller portraits 'THYROID', 'VITILIGO', 'T1DM', 'PANCREAS', 'HASHIMOTO'

WHAT IT ENCODES

PA clusters with other organ-specific autoimmune diseases — autoimmune thyroid disease (Hashimoto/Graves, the most common association), type 1 diabetes, vitiligo, Addison disease, hypoparathyroidism and is a component of autoimmune polyglandular syndrome type 2/3. Always check TSH in a PA patient and consider PA in any patient with multiple autoimmune endocrinopathies. Mnemonic: 'autoimmune travels in packs — find one (PA), screen the thyroid.'

11. Clinical: pale, glossy tongue

WHAT YOU SEE

CLINICAL face 'PALE', 'RED GLOSSY TONGUE', 'CRACKED CORNERS', 'EASY BRUISES'

WHAT IT ENCODES

Mucocutaneous signs: pallor (± lemon-yellow tint from combined pallor + mild jaundice of ineffective erythropoiesis), atrophic glossitis (smooth, beefy-red, sore tongue), angular cheilitis, and easy bruising from associated thrombocytopenia. B12 deficiency uniquely adds NEURO disease — subacute combined degeneration of the dorsal columns and lateral corticospinal tracts (paresthesias, loss of vibration/proprioception, ataxia, positive Romberg) plus cognitive change. Mnemonic: 'lemon-yellow + beefy tongue + tingling = B12.'

12. Folate: NO neuro

WHAT YOU SEE

Bottom 'FOLATE: NO NEURO', '↑MCV'

WHAT IT ENCODES

Critical discriminator: B12 deficiency (including PA) causes neurologic disease (subacute combined degeneration, neuropathy, dementia) via the methylmalonyl-CoA/myelin pathway, whereas FOLATE deficiency causes the same macrocytic anemia but NEVER the classic neuro syndrome. Therefore a megaloblastic anemia WITH neuro signs = B12 until proven otherwise, and folate must never be given alone before excluding B12. Mnemonic: 'Folate Fixes the blood but Fails the nerves — only B12 protects myelin.'

13. Juvenile / congenital IF

WHAT YOU SEE

'JUVENILE PA', baby 'CONGENITAL IF AR AGE 1-3 NORMAL MUCOSA NO ANTIBODIES'

WHAT IT ENCODES

Congenital intrinsic factor deficiency (juvenile/congenital pernicious anemia) is autosomal recessive (GIF gene), presenting at ~age 1-3 once maternal B12 stores deplete, with NORMAL gastric mucosa and NO autoantibodies — distinguishing it from adult autoimmune PA. Related inherited cause: Imerslund-Gräsbeck syndrome (cubam receptor defect → ileal B12 malabsorption + proteinuria). Treatment is the same: parenteral B12. Mnemonic: 'congenital IF = young child, normal stomach, antibody-negative — it's a gene, not autoimmunity.'